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Project Title:

Design and Implementation of Multi-Scope DHCP Services and Subnet Isolation Across a Multi-Block Residential Compound

1. Introduction and Background

Dynamic Host Configuration Protocol (DHCP) is a Layer 7 network management protocol that automates the assignment of IP addresses, subnet masks, default gateways, and DNS servers to client devices using the DORA (Discover, Offer, Request, Acknowledge) handshake process. In shared residential environments, deploying central DHCP services alongside proper network segmentation eliminates the administrative burden of manual IP assignment, prevents IP address conflicts, and organizes host traffic across subnets.

2. Problem Statement and Scope

Problem Statement:

I live in a compound with 15 rooms divided into 3 blocks, with each block containing 5 rooms. The current mobile network service in our area is extremely poor, making basic web browsing almost impossible. To solve this, our landlord suggested installing a central Starlink wifi connection where all residents contribute to the monthly subscription.

While this solves our internet access issue, it creates a major cybersecurity problem for me. As a cybersecurity enthusiast, I am not comfortable being on a flat, public network shared with 14 other households. On an unsegmented network, anyone connected to the same Wi-Fi can run packet sniffers (like Wireshark) or network mappers (like Nmap) to discover every phone, laptop, and smart device in the compound, capture unencrypted traffic, and launch lateral attacks.

To solve this, I am designing a network setup that splits our 15-room compound into 3 isolated subnets—one dedicated to each 5-room block—using a central Windows Server DHCP infrastructure and router sub-interface routing boundaries to guarantee privacy, address automation, and enforced inter-subnet firewall isolation using Extended Access Control Lists (ACLs) applied at the router sub-interface boundaries..

Scope:

This project covers the architectural design, VLSM subnetting, multi-scope DHCP deployment, and cross-subnet isolation testing for a 3-block compound topology simulated in Cisco Packet Tracers:

-WAN Boundary: Starlink operating in Bypass Mode.

- Core Router: Manages sub-interfaces, DHCP Relays (ip helper-address), and access control rules.-

Central Server: A dedicated Windows Server 2022 (192.168.10.194) serving dynamic DHCP leases for all three blocks.

3. Aim and Objectives

-Aim: Deploy a central Windows Server DHCP service to automate dynamic IP allocation across three isolated block subnets within a 15-room compound sharing a Starlink WAN link inside Cisco Packet Tracer.

- Objectives:

1. Subnet a private IPv4 block into three dedicated subnets for Block A, Block B, and Block C.
2. Configure Layer 2 VLAN access port separation (VLANs 10, 20, 30, and 40) on the Cisco 2950 switch
3. Establish an 802.1Q trunk link (Fa0/1) to connect the switch to the core router sub-interfaces.
4. Create and configure three distinct DHCP scopes with defined ranges, exclusions, lease durations, gateway, and DNS options.
5. Configure DHCP Relay Agents on the core router to relay cross-VLAN DORA broadcasts to the central server.
6. Enforce layer 3 subnet boundaries on the router to block inter-block communication while maintaining internet egress.
7. Verify automatic client lease acquisition and analyze client APIPA fallback behavior during DHCP service failure.

4. System Analysis and Design

Architecture: Centered on segmenting a 15-room residential compound into 3 logically isolated block subnets using VLSM addressing, backed by a central Windows Server 2022 DHCP infrastructure(192.168.10.194),an 802.1Q switch Trunk, and a core router.

Traffic Routing: Designed to isolate Layer 2 broadcast domains per block while relaying broadcast DORA packets across sub-interfaces to the central DHCP server via ip helper-address directives.

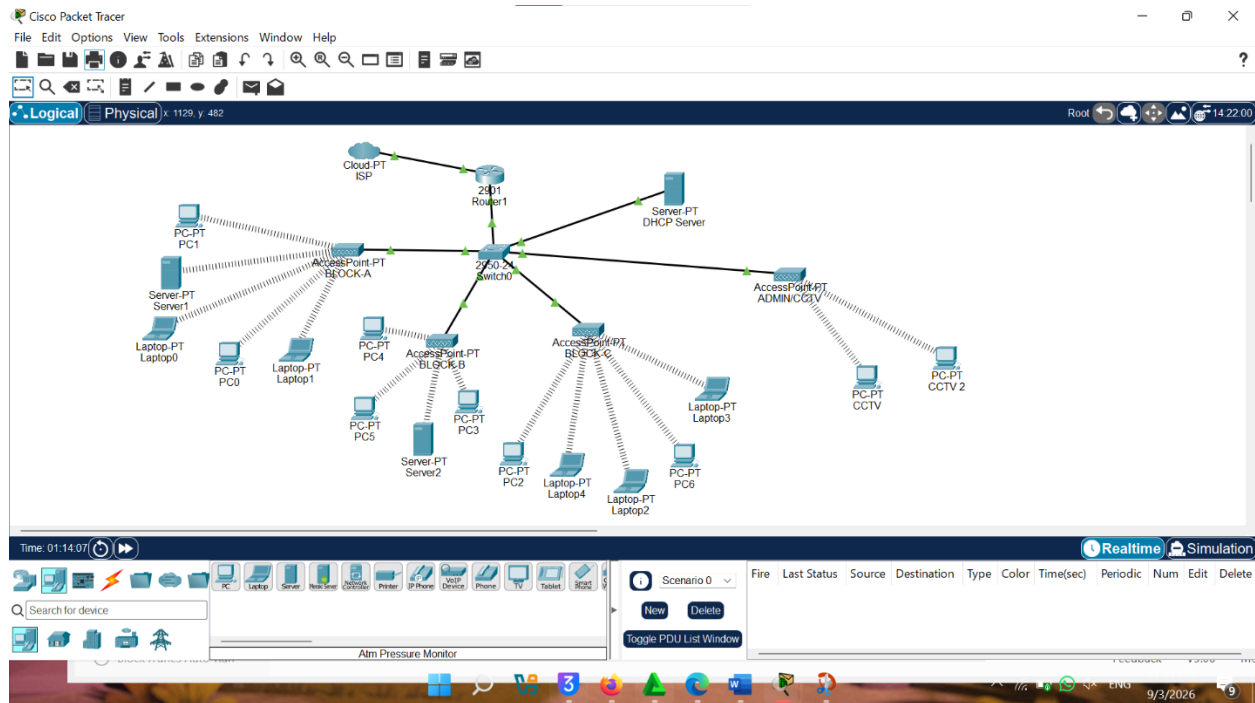


Figure 6.1: Network topology a Complete Cisco Packet Tracer logical topology showing the Starlink WAN link, Core Router, Central DHCP Server, and the three residential block subnets.

5. Tools and Materials

-Simulation Software: Cisco Packet Tracer.

-WAN Link: Starlink wifi Connection operating in Bypass Mode.

- Switching Infrastructure: Cisco 2950-24 Switch configured with access ports (VLANs 10–40) and 802.1Q trunking (Fa0/1).

-Routing Infrastructure: Cisco Core Router configured with sub-interfaces (.10, .20, .30, .40) and ip helper-address relay settings.

- Firewall / Extended ACL Implementation: Configured inbound Extended ACLs on each block sub-interface to explicitly permit UDP port 67 (DHCP forwarding to 192.168.10.194), block all cross-subnet IP traffic, and permit outbound internet traffic.

- Server Infrastructure: Packet Tracer server hosting the DHCP Server.

Testing Tools: Cisco Packet Tracer PC command prompt(ping,ipconfig).

6. Design and Implementation- Subnetting & IP Address Plan:

Divided the base network 192.168.10.0/24 using VLSM into /26 subnets (block size of 64) for Block A, Block B, and Block C, along with a /29 subnet for Admin/CCTV (192.168.10.192/29) and reserved address space for future use.

$2^2 = 4$ subnets

||||| . ||||| . ||||| . 1000000

128 64 32 16 8 4 2 1

Block size = 64

Figure 6.2: VLSM Subnetting calculation breakdown showing the base network (192.168.10.0/24), prefix expansion (/26), block size derivation (64), and IP range allocations (usable hosts, subnet masks, and broadcast addresses) for Block A, Block B, Block C, Admin/CCTV, and Reserve.

-DHCP Scope Deployment: Configured three distinct DHCP scopes on the central server matching the calculated subnets, gateway addresses, and DNS settings

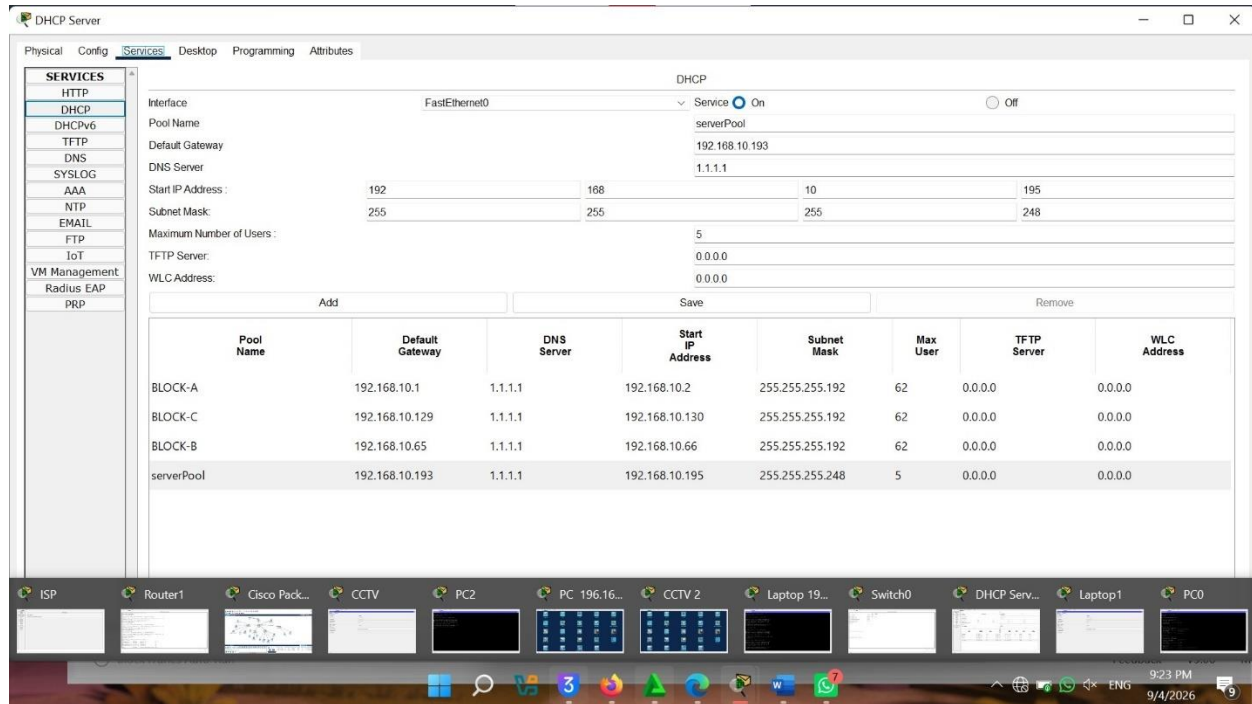
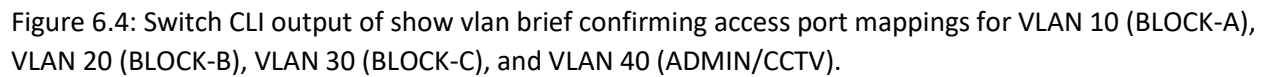


Figure 6.3: Packet Tracer DHCP SERVER Packet Tracer Server GUI demonstrating active DHCP scope configurations for Block A, Block B, Block C and serverpool representing Admin/cctv.

-Layer 2 VLAN & Access Port Assignment: Configured isolated access ports on the Cisco 2950 switch for each block AP (Fa0/2 on VLAN 10, Fa0/3 on VLAN 20, Fa0/4 on VLAN 30, and Fa0/5–Fa0/6 on VLAN 40).



The screenshot shows a Cisco IOS Command Line Interface window titled "Switch0". The interface has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The command history shows the following sequence:

```

Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch# show int trunk
Port      Mode           Encapsulation   Status        Native vlan
Fa0/1     on              802.1q          trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40

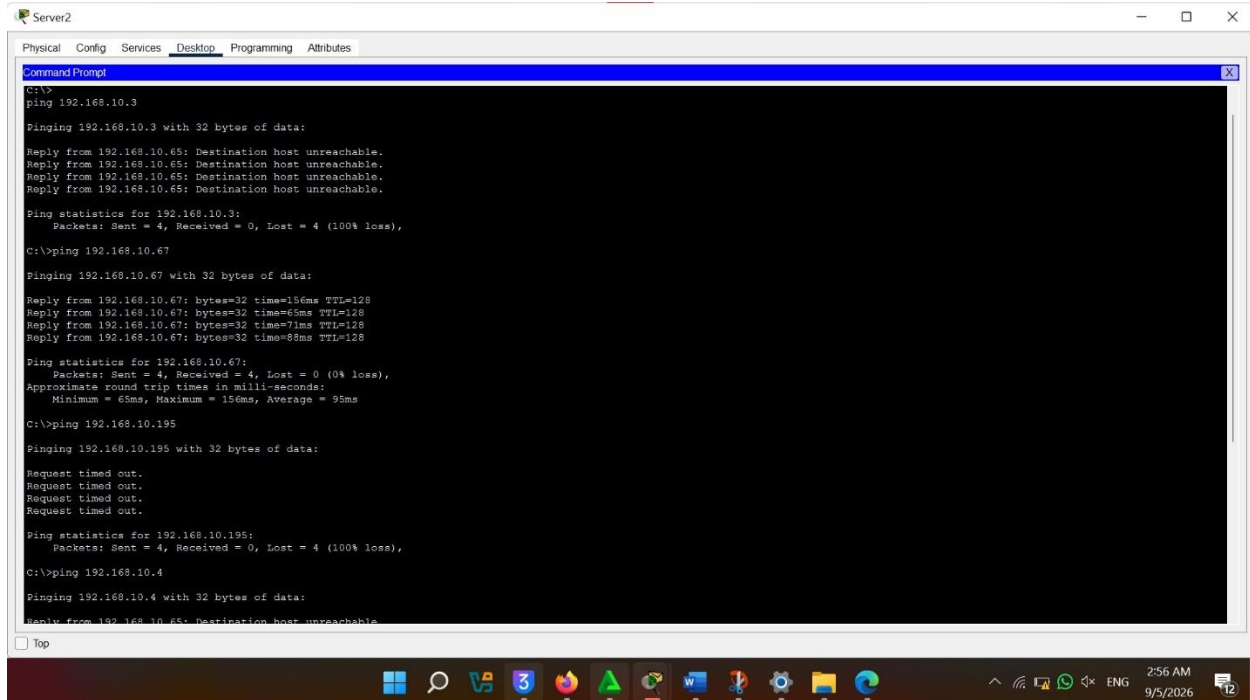
Switch#
Switch#
Switch#
Switch#
Switch#

```

At the bottom left, there is a "Top" button. At the bottom right, there are "Copy" and "Paste" buttons. The Windows taskbar at the very bottom shows various application icons and the system clock indicating 12:51 AM on 9/5/2026.

Figure 6.5: Switch CLI output of show int trunk confirming port Fa0/1 operating in 802.1q trunking mode for VLANs 1, 10, 20, 30, and 40.

- Firewall ACL Isolation Testing: Verified that ping attempts between hosts in opposing subnets (e.g., Block A 192.168.10.4 to Block B 192.168.10.66) return Destination Host Unreachable, while outbound WAN/internet connectivity remains operational.



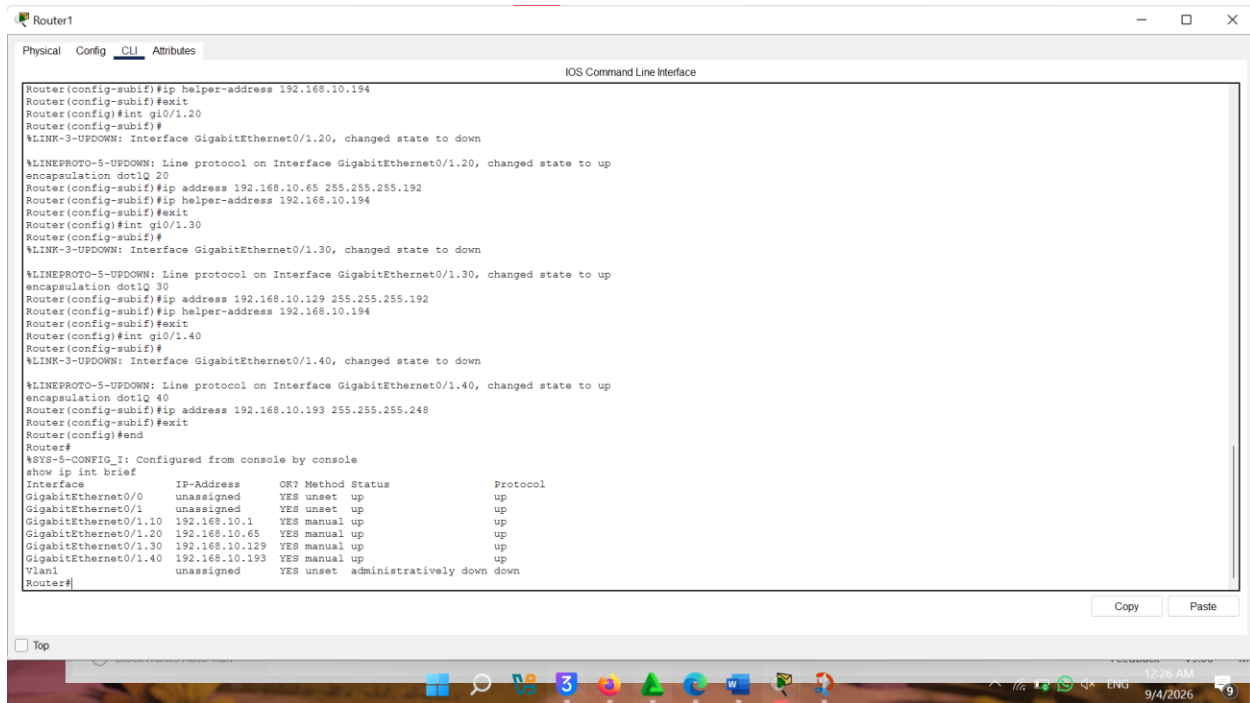
The screenshot shows a Windows Server 2012 R2 desktop environment. A Command Prompt window is open, displaying the results of several ping tests. The tests are as follows:

- ping 192.168.10.3:** Results in "Destination host unreachable" for all four attempts. Statistics: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss).
- ping 192.168.10.67:** Results in successful replies for all four attempts. Statistics: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss). Round trip times are approximately 65ms to 88ms.
- ping 192.168.10.195:** Results in "Request timed out" for all four attempts. Statistics: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss).
- ping 192.168.10.4:** The first attempt results in "Destination host unreachable".

The taskbar at the bottom shows the Windows Start button, search icon, and several application icons. The system clock indicates 2:56 AM on 9/5/2026.

Figure 6.6: a ping on the same VLAN it return receive = 4,lost = 0, it means is up while a ping from Block - B to the rest of the Blocks is returning Destination Host Unreachable or request timed out.

-DHCP Relay: Configured ip helper-address directives on the core router sub-interfaces to relay cross-VLAN DORA broadcast traffic to the central server (192.168.10.194).



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-subif)#ip helper-address 192.168.10.194
Router(config-subif)#exit
Router(config)#int gi0/1.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.10.65 255.255.255.192
Router(config-subif)#ip helper-address 192.168.10.194
Router(config-subif)#exit
Router(config)#int gi0/1.30
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up
encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.10.129 255.255.255.192
Router(config-subif)#ip helper-address 192.168.10.194
Router(config-subif)#exit
Router(config)#int gi0/1.40
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up
encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.10.193 255.255.255.248
Router(config-subif)#exit
Router(config)#end
Router#
$SYS-5-CONFIG_I: Configured from console by console
show ip int brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset up up
GigabitEthernet0/1 unassigned YES unset up up
GigabitEthernet0/1.10 192.168.10.1 YES manual up up
GigabitEthernet0/1.20 192.168.10.65 YES manual up up
GigabitEthernet0/1.30 192.168.10.129 YES manual up up
GigabitEthernet0/1.40 192.168.10.193 YES manual up up
Vlan1 unassigned YES unset administratively down down
Router#
```

Figure 6.7: Core Router CLI output showing show ip interface brief and sub-interface configuration with IP helper-address 192.168.10.2.

-Subnet Segmentation: Assigned separate sub-interfaces and VLANs per block to establish distinct broadcast domains and restrict unrouted lateral traffic.

7. Testing, Results and Discussion

-Lease Acquisition: Verified dynamic IP allocation across all block subnets via DORA propagation across the trunk line.

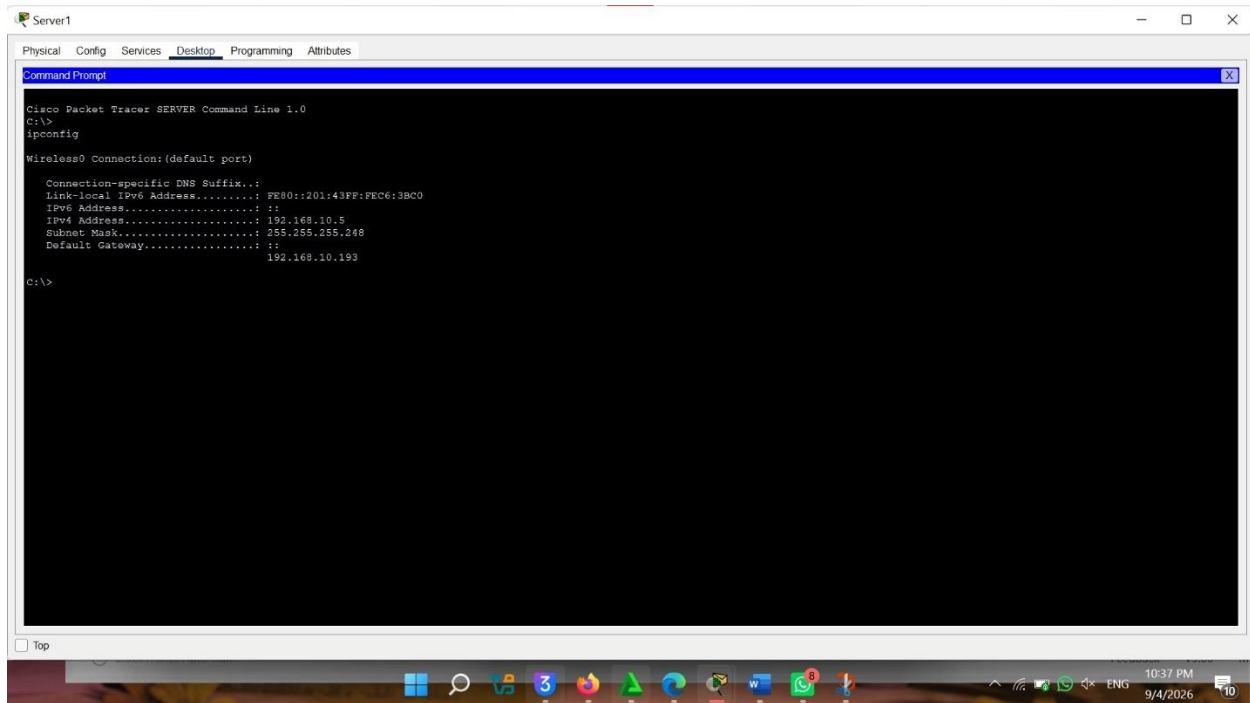


Figure 6.8: BLOCK-A Client Command Prompt showing dynamic IP acquisition via ipconfig.

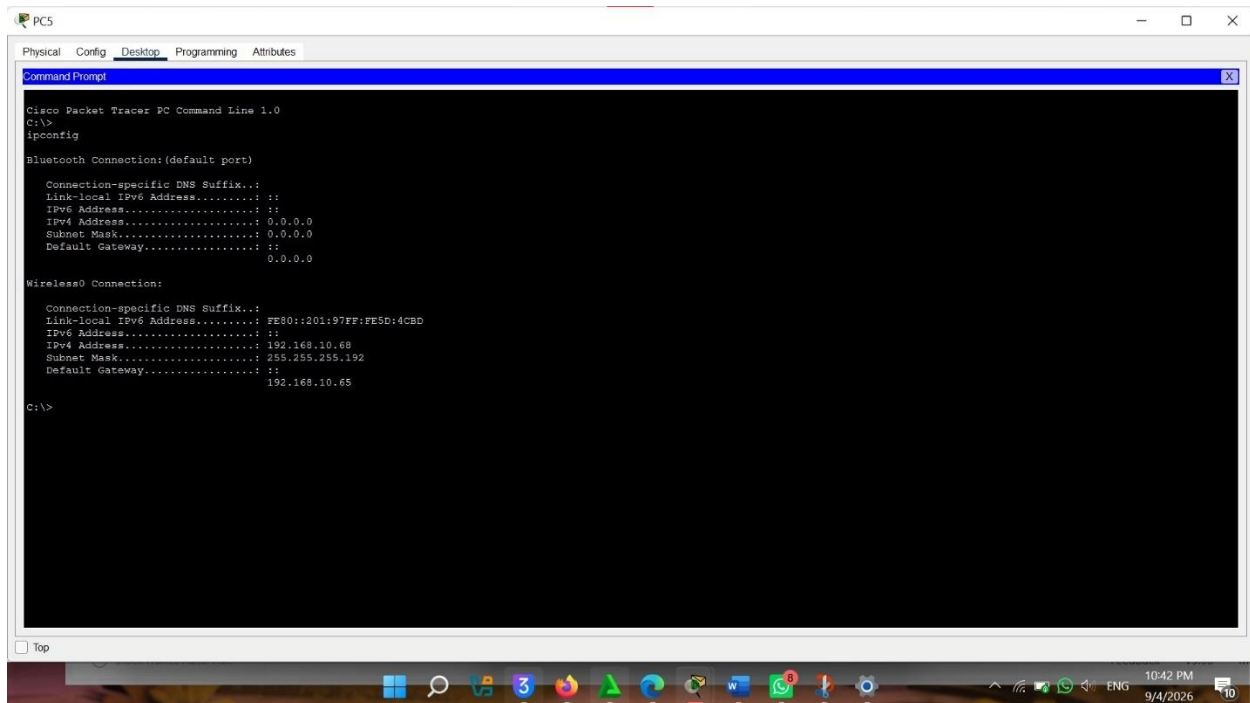


Figure 6.9: BLOCK-B Client Command Prompt showing dynamic IP acquisition via ipconfig.

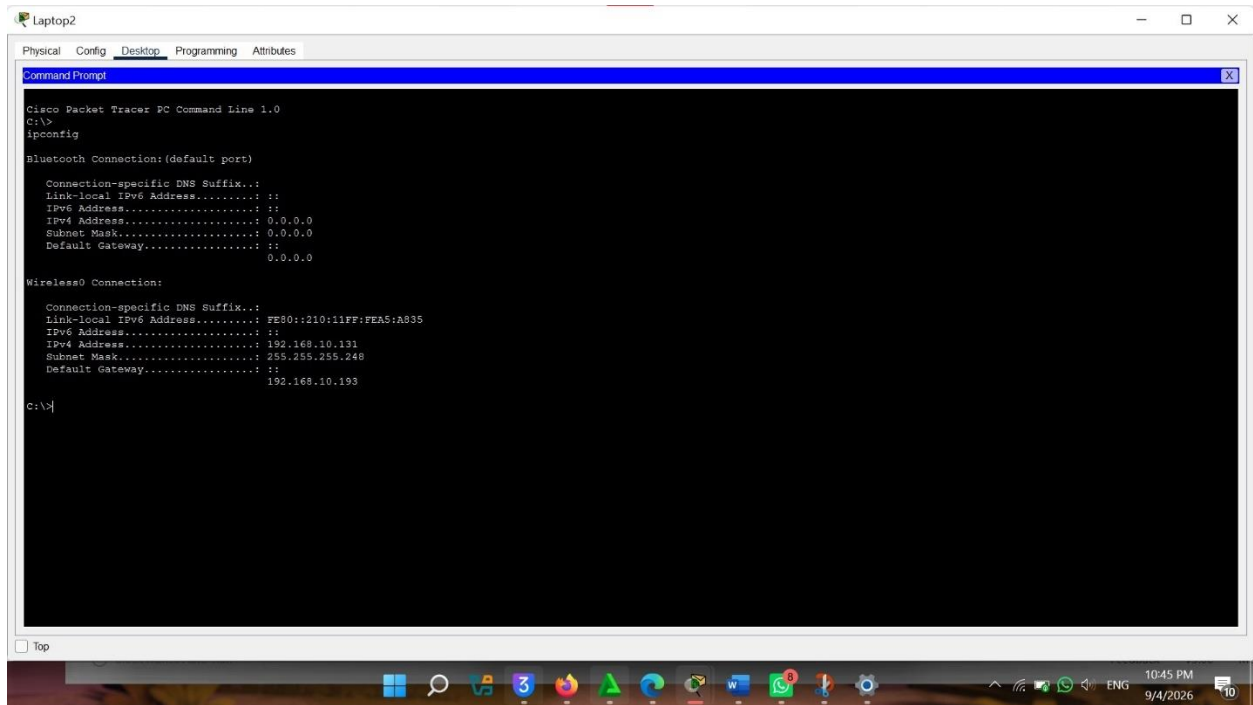


Figure 7.0: BLOCK-C Client Command Prompt showing dynamic IP acquisition via ipconfig.

-Failure Analysis: Simulated DHCP service failure by disabling the server port and verified client fallback to Automatic Private IP Addressing (APIPA).

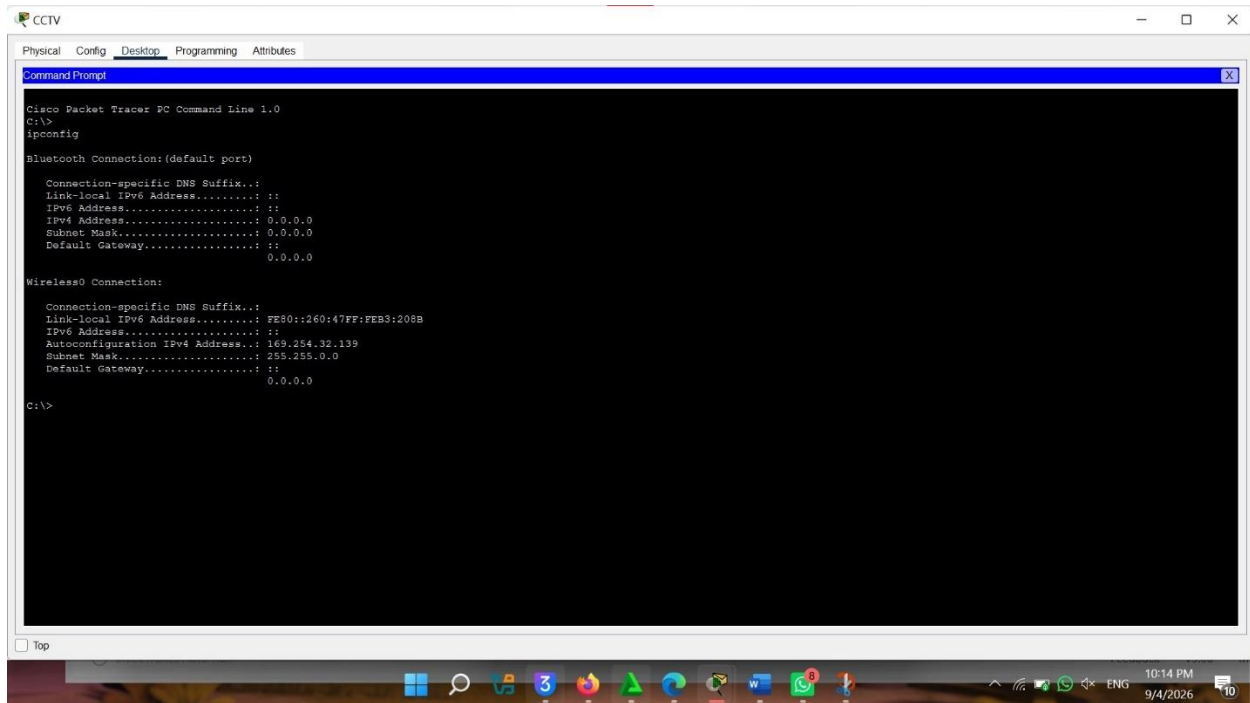


Figure 7.1: APIPA fallback output Client ipconfig output highlight that the APIPA output (169.254.32.139 with mask 255.255.0.0) confirms the failure of Layer 3 relay and DORA negotiation, causing the client OS to fall back to RFC 3927 link-local behavior.

8. Challenges and Troubleshooting

- Cross-Subnet Relay: Addressed the issue of DHCP broadcast requests failing to cross router boundaries by properly implementing ip helper-address relay settings on each block sub-interface.
- Client IP Lease Timing: Resolved client acquisition delays during service restoration by utilizing ipconfig /release and forcing STP/Wireless link convergence in Packet Tracer using Fast-Forward Time.

9. Conclusion and Recommendations

Conclusion: The project successfully implemented a central DHCP service to manage network allocations across three isolated block subnets inside Cisco Packet Tracer. By combining DHCP scope management, router relay agents, and VLSM subnet segmentation, the 15-room compound receives automated IP addressing alongside host traffic separation.

Recommendations:

- DHCP High Availability: Configure Failover (Hot Standby) on a secondary server to guarantee continuous IP leasing if the primary server goes down.
- Client Isolation on Access Points: Enable Layer 2 Client Isolation on every wireless Access Point so residents within the same block cannot scan each other's devices.

10. References

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